Study of Networks in Linux and Windows

Exp no: 1 Date:

Aim:

To study, execute and analyze various network commands used for network configuration, routing, diagnostics, and packet inspection across Window and Linux operating systems.

BASIC CROSS-PLATFORM & WINDOW NETWORKING COMMAD

Command	Protocol	Function & Description	Example command
arp -a	ARP (layer 3)	Displays the Address Resolution Protocol cache mapping target IP addresses to their MAC address	arp -a
hostname	System	Displays the system name assigned to the local computer	hostname
ipconfig/all	TCP/IP (windo	Displays detailed network configuration parameters including IP address, Subnet mask, Default gateway.	ipconfig/all
nbtstat -a	NetBios over TCP/IP	Troubleshoots NetBIOS name resolution issues by resolving names on target hosts	nbtstat -a <hostname>
netstat	TCP/UDP (layer 4)	Shows active TCP/IP connections, listening ports, interface statistics and routing tables	netstat -a
nslookup DNS (layer 7)	DNS (layer 7)	Queries Domain Name system servers to obtain domain-to-IP or MX record mappings in interactive mode	nslookup

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Command	Protocol	Function & Description	Example
pathping	ICMP/ Ping + Tracert	Windows utility that combines ping and tracert, evaluating data loss and latency per hop over 25 seconds	pathping
route	IP routing (layer 3)	Displays and manipulates the kernel/system IP routing table to direct network traffic via specific interfaces	route print

Advanced Linux Networking Commands:

A. The IP Command Suite (iproute2)

Syntax structure: ip <options> <object> <command>

1. IP Address Management

- Display active interface addresses:
  - → ip address show
- Assign an IP address to an interface (enps03):
  - → ip address add 192.168.1.254/24 dev enps03
- Delete an IP address from an interface
  - → ip address del 192.168.1.254/24 dev enps03

2. Network Management:

- Bring interface online:
  - ip link set eth0 up
- Bring interface offline:
  - ip link set eth0 down
- Enable promiscous mode:
  - ip link set eth0 promisc on

3. IP routing Management:

- Add default gateway:
  - ip route add default via 192.168.1.254 dev eth0
- Add route to specific subnet via gateway:
  - ip route add 192.168.1.0/24 via 192.168.1.254
- Add direct route via interface.
  - ip route add 192.168.1.0/24 dev eth0
- Delete a route:
  - ip route delete 192.168.1.0/24 via 192.168.1.254
- Display path taken for a specific IP:
  - ip route get 10.10.1.4

B. Legacy configuration (ifconfig)

- Usage: Traditionally used to query and configure network interfaces.
- Note: Deprecated on modern Linux systems in favor of the IP suite.

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C. Network Diagnostics (mtr - Matt's Traceroute)
mtr combines the functions of ping and traceroute into a dynamic real-time monitoring tool.

- Basic MTR query:
  - ↝ mtr google.com
- Display numeric IP addresses instead of hostnames
  - ↝ mtr -g google.com
- Display both hostnames and IP addresses:
  - ↝ mtr -b google.com
- Set specific ping count
  - ↝ mtr -c 10 google.com

D. Packet Capture and Analysis (tcpdump)

1. Installation and Interface Discovery
   - Install utility:
     - ↝ dnf install -y tcpdump
   - List available capture interfaces:
     - ↝ tcpdump -D

2. Packet Capture Execution Examples:
   - Capture all packets on interface eth0:
     - ↝ tcpdump -i eth0
   - Limit capture to 10 packets:
     - ↝ tcpdump -i eth0 -c 10

- Filter by network subnet
  - tcpdump -i eth0 net 10.10.0.0/24

4. ICMP diagnostics Error Analysis (ping)

When performing connectivity tests using ping, responses indicate network conditions and failure modes:

- ping google.com
- ping -c 10 google.com

Analysis of Ping Error Messages:

- Destination Host Unreachable:
  - Cause: The host or an intermediate router lacks a route the largest destination target.
- Request timed out:
  - Cause: No ICMP Echo reply was received within the default timeout period.
- Unknown host:
  - Cause: Name resolution failed. The system should not resolve the specified hostname to an IP address due to misspelled hostnames or misconfigured DNS servers.